

Group A
[Answer all the questions]

1. Answer any FIVE

5x1=5

- a) What is Regular Language?
- b) What is a Finite Automata?
- c) Draw the DFA for the string ends with a.
- d) What is proof by counter-example?
- e) What is PushDown Automata?
- f) Write the RE for an odd length string.
- g) What is an escape sequence?

2. Answer any FOUR

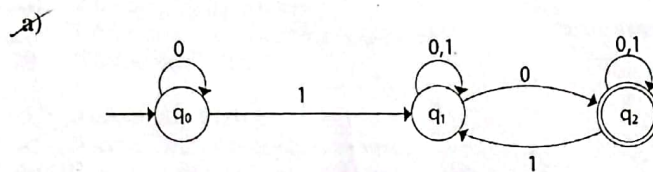
4x2.5=10

- a) Show that the following languages are not regular: $L = \{a^{n^2} \mid n \geq 1\}$
- b) "Every DFA is a NFA but not vice versa" - explain that statement.
- c) Remove null production from the following grammar.
 $S \rightarrow ABAC, A \rightarrow aA| \epsilon, B \rightarrow bB| \epsilon, C \rightarrow c$
- d) Write the RE that will recognize only the registration number of the student of SUST.
- e) What is the Pumping Lemma for Regular language? Using Pumping Lemma, prove that the language $A = \{a^n b^n \mid n \geq 0\}$ is not Regular.
- f) Prove the following equality by induction for all $n \geq 0$:

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

3. Answer any TWO

2x5=10



Convert the given NFA to DFA.

- b) Give state diagrams of DFAs recognizing the following languages. In all parts, the alphabet is $\{0,1\}$.
 - a. $\{w \mid w \text{ is any string except } 11 \text{ and } 111\}$
 - b. $\{w \mid \text{every odd position of } w \text{ is a } 1\}$
- c) Let $S(n) = 1 + 2 + \dots + n$ be the sum of the first n natural numbers and let $C(n) = 1^3 + 2^3 + \dots + n^3$ be the sum of the first n cubes. Prove the following equalities by induction on n :
 - a. $S(n) = \frac{1}{2}n(n+1)$
 - b. $C(n) = \frac{1}{4}n^2(n+1)^2$

Group B

[Answer all the questions]

5x1=5

4. Answer any FIVE

- Write the RE for an even length string only.
- What is the purpose of wildcards in RE?
- What is proof by induction?
- What is a Turing machine?
- Give an example of proof by counterexamples.
- Draw the DFA for the string ends with b.
- What is CFG?

4x2.5=10

5. Answer any FOUR

- Remove unit production from the following grammar.
 $S \rightarrow XY, X \rightarrow a, Y \rightarrow Z|b, Z \rightarrow M, M \rightarrow N, N \rightarrow a$
- Why does the Finite Automata can't solve the counting problem but the PDA can?
- Write regular expressions for the following languages over the alphabet $\Sigma = \{a, b\}$:
 - All strings that do not end with aa.
 - All strings that contain an even number of b's.
- Construct a DFA over $\Sigma := \{a, b\}$ that accepts the following language: $\{w \in \Sigma^* \mid \text{each } a \text{ followed by exactly 1 or 3 b's}\}$
- Show that the following languages are not regular: $L = \{a^p \mid p \text{ prime number}\}$
- Give the formal definition of a Turing Machine.

2x5=10

6. Answer any TWO

- Draw the state diagram of TM that recognize the following Language,
 $L = \{a^n b^{2n} \mid n \geq 1\}$
- Draw the PushDown Automata for the language $L = \{0^a 1^b 0^c \mid b = a + c\}$
- Write the formal definition of NFA.
 Give state diagrams of NFAs with the specified number of states recognizing each of the following languages. In all parts, the alphabet is $\{0, 1\}$.
 - The language $\{w \mid w \text{ ends with } 00\}$ with three states
 - The language $0^* 1^* 0^+$ with three states

1+4

Shahjalal University of Science and Technology
Institute of Information and Communication Technology (IICT)
Software Engineering
2nd Year 2nd Semester Final Examination 2023 (Session: 2020-21)
Course Code: SWE 235 Credits: 2 Course Title: Ethics and Cyber Law
Time: 2 hrs Total Marks: 50

Group A

[Answer all the questions]

1. Answer any FIVE 5x1=5
- a) When is a person's behavior considered ethical?
 - b) What is Ethics?
 - c) What do you mean by plagiarism?
 - d) Is stealing ALWAYS wrong? When is it right to steal?
 - e) What do understand by the term internet privacy?
 - f) What does integrity mean to you?
2. Answer any FOUR 4x2.5=10
- a) How Ethics fit into engineering?
 - b) What should you do in following case? Why?
You are supervising a product with specifications that only U.S.-made parts may be used. Late in the project you discover a sub-contractor has supplied a part with foreign-made bolts. They aren't very noticeable and would function identically to U.S.-made bolts. Your customer urgently needs the finished product.
 - c) What is dilemma in Ethics?
 - d) What is Utilitarian Approach? Explain with example.
 - e) Write short notes on Cybercrime against individuals.
3. Answer any TWO 2x5=10
- a) If you want to set up a nuclear power plant, what would be the reasoning or arguments for and against that decision making?
 - b) Imagine you are a software developer working for a company that has asked you to create a program that collects user data for marketing purposes. The data collection is legal, but it raises ethical concerns. What ethical considerations should you take into account when deciding whether to work on this project?
 - c) Discuss the crimes against society by using information communication technology.

Group B

[Answer all the questions]

4. Answer any FIVE (Is it ethical or unethical? Why?) 5x1=5
- a) Extrajudicial killing
 - b) Death penalty
 - c) Recruiting your friend in your company
 - d) Who are grey hat hackers?
 - e) Taking credit for your team's research
 - f) What is identity theft?
5. Answer any FOUR 4x2.5=10
- a) What is the difference between ethics and morality?
 - b) Can an action be morally right but ethically wrong, or vice versa? Provide an example.
 - c) What are the reasons that Engineers should follow their professional code of ethics?
 - d) What is the cyber terrorism? Give an example of this kind of crime.
 - e) What is the punishment for Punishment for Any propaganda or campaign against liberation war, Cognition of liberation war, Father of the nation, National Anthem or National Flag.
6. Answer any TWO 2x5=10
- a) What are the ethical issues that engineers encounter? Explain with examples.
 - b) What is webcam hacking? What are the steps one should follow to be safe from webcam hacking?
 - c) Define cyber terrorism. Briefly describe the punishment of cybercrimes in Bangladesh.

Part A

1. Answer the following Questions (Any Five). 5 × 2 = 10
- (a) What is the time complexity for 'Sieve of Eratosthenes'?
 - (b) What do understand by open-address hashing? Give an example.
 - (c) What is phi Function and why it is used?
 - (d) Describe a scenario when you will use 'Segmented Sieve'.
 - (e) What is 'Linear Diophantine Equation'? Write the general form of the equation.
 - (f) What is the mathematical definition of CRT?
 - (g) What is the time complexity of 'Extended Euclidean Algorithm'?
2. Answer the following Questions (Any Four). 4 × 5 = 20
- (a) Calculate the GCD(-72, 5) using the Euclidean Algorithm.
 - (b) Modify the pseudo-code for the extended euclidian algorithm for generating more (x,y) which satisfies the equation that extended euclidian solves.
 - (c) Use the bigmod algorithm to find $36^{7789} \bmod 13$.
 - (d) You can use one of Gaus theorem to calculate $\phi(n)$. State that theorem & write the pseudo-code to find the totient from 1 to n.
 - (e) How you can find the smallest x where x satisfies the following condition? Here m_1 and m_2 are not co-prime.
$$x \equiv a_1 \pmod{m_1}$$
$$x \equiv a_2 \pmod{m_2}$$
3. Answer the following Questions (Any TWO). 2 × 10 = 20
- (a) Write down the approach and pseudo-code for the following problem.
You will be given a number N, output the number of factors of N.
 $1 \leq N \leq 10^{17}$.
 - (b) Derive the equation to find the value 'x' in the Chinese Remainder Theorem for non co-prime module.
 - (c) Write down the code for the following problem(with explanation in comments) & what approach will you follow.
Given two integers A and B, find number of primes inside the range of A and B inclusive. Here, $1 \leq A \leq B \leq 10^6$ and the runtime for the algorithm should be $O(B - A)$.

Part B

4. Answer the following Questions (Any Five). 5 × 2 = 10
- (a) What is the special benefit of the bellman ford algorithm?
 - (b) What is the difference between the "Euclidean" and the "Extended Euclidean" algorithms?
 - (c) What is stable sorting?
 - (d) What is the time complexity for Matrix multiplication using dp?
 - (e) What is the lps array stands for in KMP?
 - (f) What types of problems can be solved by 0-1 knapsack?

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Part A

1. Answer the following Questions. (Any Five)

5 × 2 = 10

- (a) What is the Kernel in OS?
- (b) What facts should you keep in mind while designing an OS?
- (c) Define the types of OS Organizations with examples.
- (d) What is Context switching?
- (e) What is "Timeslice" for a process?
- (f) What is "Thread"?
- (g) What is "mutex lock" and why is used for?

2. Answer the following Questions. (Any Four)

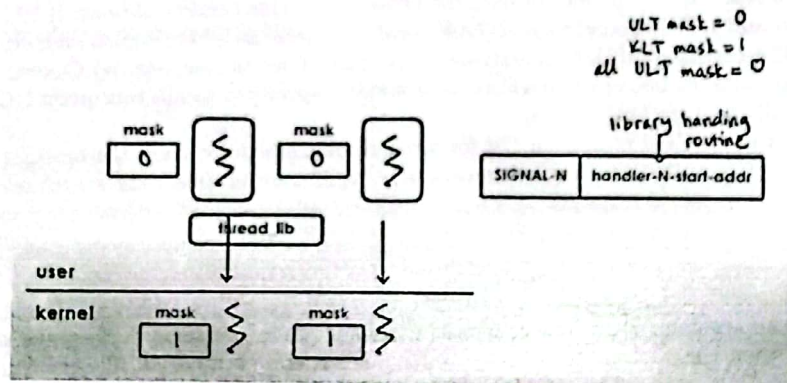
4 × 5 = 20

- (a) What do you understand by "User-Kernel Protection Boundary"? Explain and illustrate a system call from the user level.
- (b) What information holds PCB for a process? Suppose you have P_1 (running) and P_2 (queued) process. What happens to the PCB when the OS decides to switch the running process to P_2 ?
- (c) Suppose you have two systems. For system A, the relation between timeslice (TP) and scheduling time(TS) is $TP = TS$. For system B, this relation is $TP = 10 * TS$. Which system is better if you choose the matrix "Useful CPU Work" ?
- (d) Describe the "Shared Memory Multi-processor" architecture.
- (e) How do you know if a task is CPU or I/O intensive while scheduling a CPU with different timeslice values for different tasks?

3. Answer the following Questions. (Any Two)

2 × 10 = 20

- (a) How the system will handle the signal produced by the threads in the following case.



- (b) An image web server has three stages with average execution times as follows: I) Stage 1: read and parse request (10ms), II) Stage 2: read and process image (30ms), III) Stage 3: send image (20ms). You have been asked to build a multi-threaded implementation of this server using the pipeline model. Now, answer the following questions

- [i] How many threads will you allocate to each pipeline stage?
- [ii] What is the expected execution time for 100 requests (in sec)?
- [iii] What is the average throughput of this system (in req/sec)? Assume there are infinite processing resources (CPU's, memory, etc.).

- (c) Imagine you are developing a system that simulates the interaction between producers and consumers sharing a bounded buffer (can contain max N items). The producers generate items, and the consumers consume these items from the buffer. The buffer has a limited capacity, and both producers and consumers need to coordinate to avoid overflows or underflows. And they are acting on different threads. Now design the pseudo-code to add and consume from the buffer with proper mutex lock so that the above system can maintain consistency.

Part B

$$5 \times 2 = 10$$

4. Answer the following Questions. (Any Five)

- (a) What is "Hyper-Threading"?
- (b) What is "Deadlock"?
- (c) What is MLFQ?
- (d) Define "Hot Cache" and "Cold Cache".
- (e) What is 'light process state'?
- (f) What is cache affinity?
- (g) What is race condition?

$$(c) \frac{6}{6000} =$$

$$5 \times 4 = 20$$

5. Answer the following Questions. (Any Four)

- (a) Explain the ways that the OS maps the user-level thread to the kernel level. Also, define the pros and cons for each.
- (b) What are the challenges in programming in a multicore system?
- (c) For the 10-step toy order application, we have designed 2 solutions: (1) a boss-workers solution and (2) a pipeline solution. Each solution has 10 threads. In the boss-workers solution, a worker processes a toy order in 120ms. In the pipeline solution, each of the stages (= step) takes 30 ms. How long will it take for these solutions to complete 103 toy orders?
- (d) What are the advantages of having multiple DS instead of a single PCB? Define what information should be contained in a basic 'kernel-level' scheduler from the perspective of using multiple data structures.
- (e) How does hyperthreading hide memory latency?

$$2 \times 10 = 20$$

6. Answer the following Questions. (Any Two)

- (a) On a single CPU system, consider the following workload and conditions: i) 10 I/O-bound tasks and 1 CPU-bound task, ii) I/O-bound tasks issue an I/O operation once every 1 ms of CPU computing, iii) I/O operations always take 10 ms to complete, iv) Context switching overhead is 0.1 ms, v) I/O device(s) have infinite capacity to handle concurrent I/O requests, vi) All tasks are long-running.
What is the CPU utilization (%) for a round-robin scheduler where the timeslice is 20 ms? [Hint: $\text{Cpu utilization} = \frac{\text{cpu_running_time}}{(\text{cpu_running_time} + \text{ctx_switch_overheads})} \times 100$]
- (b) Suppose you have 2 I/O bound tasks arriving in the CPU scheduler at the same time. Both have exec. time of 10s, ctx. switch time = 0.1s, I/O operation is issued every 1 sec and completes I/O in 0.5s.
As you are fixing the timeslice for the scheduler you are confused between the value 1s or 5s. What timeslice you should choose for this scenario? Mathematically explain.
- (c) Consider two CPU-bound tasks simultaneously arriving in the CPU scheduler, each with an execution time of 15 seconds and a context switch time of 0.2 seconds. While determining the timeslice for the scheduler, you are uncertain whether to choose a value of 1 second or 5 seconds. Provide a mathematical explanation for the optimal choice in this scenario.

Shahjalal University of Science and Technology
Institute of Information and Communication Technology (IICT)
Software Engineering
2nd Year 2nd Semester Final Examination' Dec 2023 (Session: 2018-19)
Course Code: SWE 237 Credits: 2 Course Title: Management Information System
Time: 2 hrs Total Marks: 50

Group A
[Answer all the questions]

5x1=5

1. Answer any FIVE

- a) What is an information system?
- b) What are disruptive technologies?
- c) What is data mining? What is the difference between query and data mining?
- d) Define Moore's law.
- e) Define packet switching.
- f) What defines phishing?
- g) What does the Golden Rule signify in ethical conduct?

4x2.5=10

2. Answer any FOUR

- a) How does Information Technology (IT) influence the economic impact of a firm through the lenses of Transaction Cost Theory and Agency Theory?
- b) How does cloud computing revolutionize accessibility to IT resources and applications for businesses, and what are the primary concerns and advantages associated with this technological shift?
- c) Elaborate on the operational mechanics of a DDoS attack and how it functions?
- d) Discuss the mechanism of public key encryption.
- e) Discuss the differences between pull-based supply chain model and push-based supply chain model
- f) How information technology improves business processes?

3. Answer any TWO

2x5=10

- a) As the owner of a retail company with Sales, Inventory, Finance and Human Resources departments, how does the Management Information System (MIS) effectively integrate transactional data from these disparate Transaction Processing Systems (TPS)?
- b) How do databases address the limitations of traditional file systems, and what specific features or functionalities do databases offer to resolve these issues?
- c) What distinguishable features characterize Web 2.0 and Web 3.0 in their evolution?

Group B

[Answer all the questions]

5x1=5

4. Answer any FIVE

- a) What is your understanding of Open-Source Software?
- b) What does the Ethical "No Free Lunch" Rule mean?
- c) What is Grid Computing?
- d) Define Big Data.
- e) What are Trojan horses?
- f) What is the churn rate in business?
- g) How does Metcalfe's Law define the value of a network within the context of network economics?

4x2.5=10

5. Answer any FOUR

- a) What is Customer Relationship Management? What are the goals of a CRM system?
- b) Elaborate the concept of Service-Oriented Architecture (SOA).
- c) How do spoofing programs differ from sniffer programs in terms of their functionalities and operations?
- d) What are the challenges posed by enterprise applications?
- e) What are the features of the Agile software development process?
- f) What are the pros and cons of utilizing system prototyping in product development?

2x5=10

6. Answer any TWO

- a) Imagine you're part of a team that has unearthed a critical security flaw in one of the company's flagship products. This vulnerability, if exploited, could potentially lead to severe data breaches and significant harm to the product's users. How would you approach this situation using structured ethical analysis steps?
- b) Discuss on the functionality and operational mechanism of DNS (Domain Name System).
- c) In a scenario where Bob sends a message to Alice over the internet, how do they ensure that the message remains unaltered and verify the authenticity of the sender, preventing unauthorized manipulation or sender imposters? Describe the method

Shahjalal University of Science and Technology

Institute of Information and Communication Technology

BSc (Engg.) In Software Engineering

2nd Year 2nd Semester Final Examination 2022

Course: Numerical Analysis (SWE 231)

Credits: 2.0 Full Marks: 50 Time: 2 Hours

[Answer every question]

Group A

Q.1 Answer any 2 questions. [2 x 2.5]

- Do you understand how numbers are represented in digital computers and how this representation induces round-off error? In particular, Do you know the difference between single and extended precision?
- Do you recognize the difference between true relative error ϵ_t , approximate relative error ϵ_a , and acceptable relative error ϵ_s ? Also show how ϵ_a and ϵ_s are used to terminate an iterative computation.
- Explain limitations of Bisection method.
- Compare the False position method and Secant method for root finding.

Q.2 Answer any 2 questions. [2 x 10]

✓ a) Answer the following. [4+6]

- Discuss the method of False Position to find an approximate root of an equation $f(x) = 0$.
- Find the real root of the equation $x^3 - 2x - 5 = 0$ by using Newton-Raphson's method (use $a = 2$ and $b = 3$).

✓ b) Answer the following. [4+6]

- Derive Lagrange's interpolation formula for unequal intervals.
- Using Lagrange's interpolation formula, find the form of the function $f(x)$ from the following table.

x	0	1	3	4
$f(x)$	-12	0	12	24

c) Answer the following. [5+3+2]

- Solve, by Euler's method of the equation $\frac{dy}{dx} = x + y$, $y(0) = 1$, for $0 \leq x \leq 1$ taking $h = 0.1$.
- Find the Taylor series expansion of $f(x) = e^{2x+1}$ at $a=0.5$ (the center of expansion).
- How large can the error be when $(n+1)$ terms are used to approximate e^{2x+1} with $x=1$?

Group B

Q.3 Answer any 2 questions. [2 x 2.5]

- ✓ Compare the performance of Gauss-Jacobi and Gauss-Seidel iterative method for solution of system of linear equation.
- Write the differences between least-square regression and interpolation.
- ✓ Between Trapezoidal rule and Simpson's $1/3^{rd}$ rule, which gives less error during integration? Why?
- d) Given 5 data points (3,4), (-1,-2), (7,9), (0,0) and (4,3) - find the sum of residuals, $\sum \epsilon_i$, where $\epsilon_i = y - y_{predicted}$ using $x = y$ as regression curve.

Q.4 Answer any 2 questions. [2 x 10]

✓ a) Answer the following. [4+6]

- Deduce the Trapezoidal rule for the numerical integration.
- ✓ Calculate an approximate value of $\int_{-3}^3 x^4 dx$ by taking seven equidistance ordinates, using the A) Trapezoidal rule B) Simpson's $1/3$ rule.

✓ b) Answer the following. [5+5]

- Discuss 'Gaussian Elimination' method for solving the system of linear equations.
- Solve the following system of linear equation by the Gaussian Elimination method.
 $2x + 2y + z + 2u = 7$
 $x - 2y + 0z - u = 2$
 $3x - y - 2z - u = 3$
 $x + 0y + 0z - 2u = 0$

c) Answer the following. [5+5]

- Find the first, second and third derivatives at the point $x = 1.5$ of the function $y = \sqrt{x}$ tabulated below.

x	1.00	1.05	1.10	1.15	1.20	1.25	1.30
$y = \sqrt{x}$	1.00000	1.02470	1.04881	1.07238	1.09544	1.11803	1.14017

- Using Newton's formula for Interpolation, estimate the population for the year 1905.

Year	1891	1901	1911	1921	1931
Population	98752	132285	168076	195690	246050